

1 Team-up and Proposal

- **Project team-up submission is due April 19th.** Each project should have 4-5 contributors. For a team of fewer than or equal to 3 members, please speak to the instructor during the offline exercise session. Start forming your team early on. And, no worries if you do not have a team, we will help you team up with other students.
- [10 pts] **The project proposal is due May 1st.** Provide a brief description of the project at most 2-page, excluding the references. Summarize the project, explain related works, and state directions to be explored. **Please plan ahead.**
- You will receive feedback from the teaching crew on the proposal.

2 Milestone Report

- [20 pts] **The milestone report is due May 17th.** The milestone report should describe progress since the proposal and the preliminary results. The length should be 2-3 pages excluding the references.
- You will receive feedback from the teaching crew on the progress.

3 Presentation and Final Report

- [20 pts] **The final project presentation** will be held on **June 12th**, the last class of the course. We will have poster presentations for all teams.
- [50 pts] **The final report is due June 19th.** The main body of the report should be 4-5 pages excluding the references and the appendix. In the appendix, you can include up to 6 pages of supplementary material that contains the details of the project if you would like. See the section “Writing a final report” on page 2 for tips on the final report.

Tips on Choosing a topic

You are welcome to pick any topic for the project you wish so long as there is some direct connection to reinforcement learning. For your project, you can apply RL methods to a new application, implement/experiment RL algorithms, or you can think about a theoretical problem or algorithm, or you can do a blend of both. Working as a team is very much encouraged – you are likely to learn more, have more fun, and accomplish more if working with fellow classmates. Here are examples of possible types of projects:

- Design algorithms and run experiments on some simulation environments like advanced environments on openAI gym, Mujoco simulator. Address some of the challenging aspects of RL, e.g., exploration, hierarchical RL, off-policy learning.
- Build or apply an algorithm tailored to a particular application coming from some other fields, such as healthcare, recommender systems, dynamic pricing, natural language processing, (simulated) robotics, etc.
- Work on a theory RL problem, and attempt to make progress on it. Besides the discussions in published papers, a great place to look for these is the proceedings of recent COLT conferences (Conference on Learning Theory) or theory papers at recent ICML and NeurIPS conferences. If you are interested in this, please speak with the instructor.
- Come up with your own idea! Be original and creative.

Writing a Final Report

The end result of your project should be a written report clearly and concisely describing what you did, what results you got and what the results mean. Your report should be 4-5 pages long. If necessary, you can include further details or plots/figures in at most 6 page appendix. The report should use 11pt font, 1-inch margins, and single spacing. Papers that vary from these guidelines risk receiving a grade deduction and/or some sections not being read.

Your report should follow the general format of a scholarly paper in this area. You should write your report as clearly as possible in a manner that would be understandable to a fellow student of this course. In other words, you should not assume that the reader has background beyond what has been covered in class (as well as the prerequisites of the course). Your report should begin by describing the problem you are studying, some background (what has been done before) and the motivation for the problem, i.e., why it is worth studying. Previous work and outside sources should be cited throughout your report in a scholarly fashion following the style of academic papers in this area.

Next, you should clearly explain what you did, both from a high level, and then in more detail. For an experimental paper, you should explain the experiments in enough detail that there is a reasonable possibility that a motivated reader would be able to replicate them. You also should outline some of the theory underlying the algorithms you are studying.

State your results clearly, and think about graphical tools to make your results clearer (a table of numbers is usually less compelling than a graphical representation of the same data). Look through published papers for ideas. For a theoretical paper, the model and other mathematical details should be explained well enough for the results to be stated with mathematical precision and clarity.

In every case, be sure to explain the meaning of your results. Don't just give a table of results or a dry mathematical formula. Explain what the results mean, and what conclusions can be drawn from them. Again, do all this in a way that would be understandable and interesting to a fellow student of this course. What did you expect to find? What did you find instead? What are the implications? If you found something surprising, can you think of how it might be explained? What you will be graded on: Projects will be graded along the following dimensions:

- originality and creativity
- experimental design and execution and/or theoretical component
- discussion and interpretation of results
- writing of the final report, including clarity, completeness and conciseness
- overall effort